

**RAJIV GANDHI COLLEGE OF ENGINEERING AND
TECHNOLOGY**

COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT - 4

Prepared by: Mohamed B Sirajuddeen

Reg No : 22TD0053

Subject Name : Web Technology

Subject Code : CST63

Year/Sem/Sec : Y3 / S6 / B

Date of Submission: 21 / 4 / 2025

Unit : 4

Teacher's Sign :

Department Vision And Mission

Vision

Cultivate student as technocrats, researchers and entrepreneurs in the field of computer science and engineering.

Mission

- To impart quality education in Computer Science and Engineering by means of learning techniques and value-added courses.
- To inculcate professional excellence and research culture by encouraging projects in cutting-edge technologies through industry interaction.
- To build leadership skills, ethical values and teamwork among the students.
- To strengthen the collaboration of department and industry through internships, mentorships and professionals body activities.

PART - A

1.) Define multimedia.

- * Multimedia is the integration of multiple forms of media such as text, audio, images, video and animation into a single interactive presentation or Application.
- * It is commonly used in education entertainment and communication.

2) Strategies of E-commerce security.

- * Data encryption
- * Secure payment Gateways
- * Strong Authentication.
- * Firewall and Antivirus protection.
- * Regular security Audits.
- * User Access control
- * Secure hosting environment

3) Define web crawling.

- * Web crawling is a process by which automated programs called web crawlers or spiders, systematically browse the internet to collect and index information from website.

- + This data is then used by search engine to update their indexes and provide relevant search results.

4) Audio and video speech synthesis.

Audio speech synthesis: The use of Text to Speech (TTS) tool to convert text on a website into spoken Audio.

Video speech synthesis: The generation of Talking avatars or lip-synched video that speak the given text.

used in virtual assistants, tutorials and interactive web content.

5) Explain online E-payment system,

- * Credit card payments
- * Debit card payments
- * Digital wallets (Eg. paypal, google pay etc)

PART - B

1) Explain N-tier Architecture and its components.

* In software engineering, multitier architecture or multilayered architecture is a client-server architecture in which presentation, Application processing and data management functions are physically separated.

* N-tier Application architecture provides a model by which developers can create Flexible and reusable application.

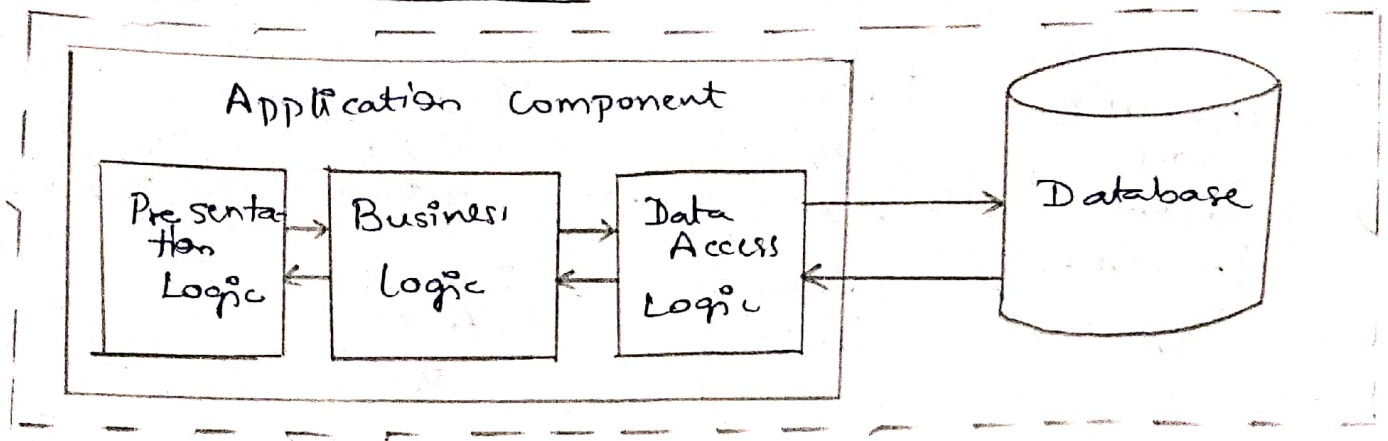
A Three-tier architecture composed of.

- Presentation Layer / Tier
- Business / Logic Tier
- Data storage Tier

Common Layers:

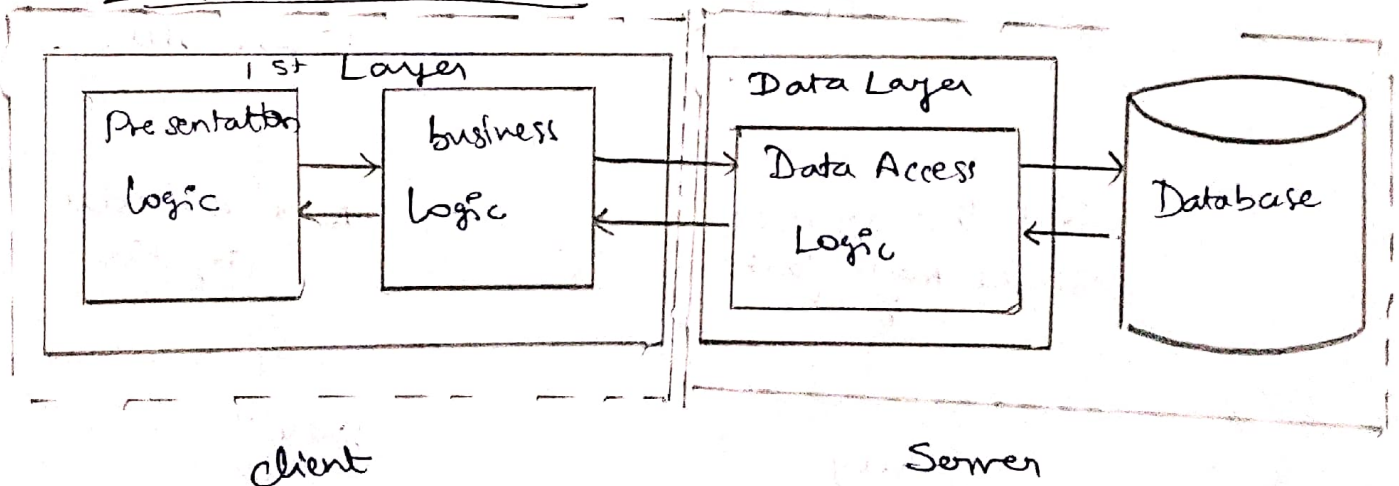
- * Presentation Layer (UI Layer, view Layer, presentation Time in multi tier architecture)
- * Application layer (business logic Layer, domain layer).
- * Data Access layer (persistence layer, Logging, networking)

1 - Tier Architecture



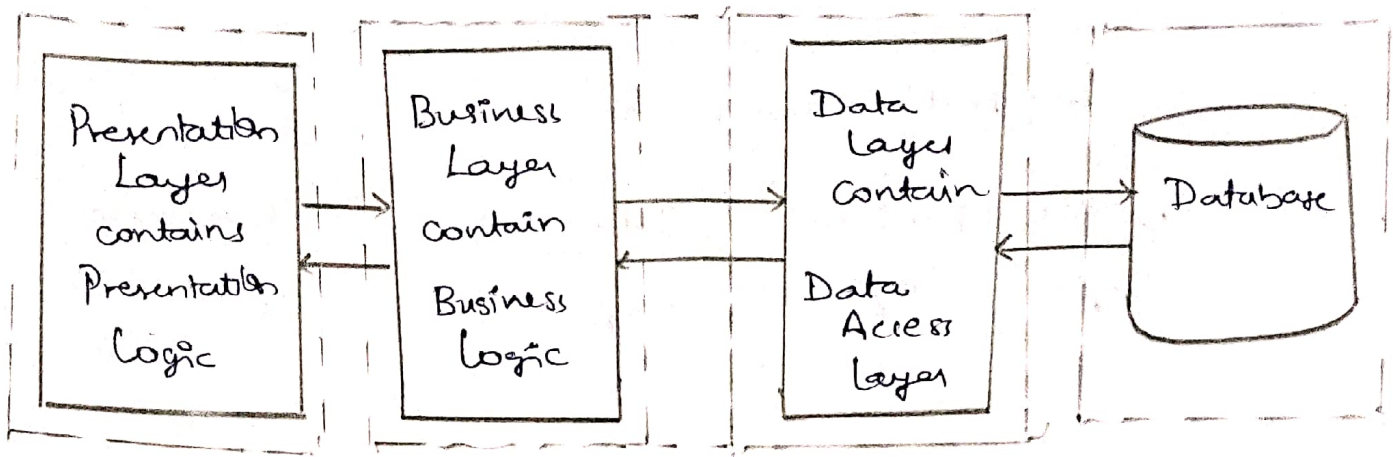
- All 3 layers are on the same machine.
- Presentation Logic, Data layers are tightly connected.
- Scalability: Single processor means hard to increase volume of processing.
- Portability: Moving to a new machine.
- Maintenance: Changing one layer requires changing other layers.

2 - Tier Architecture



- Database runs on server separated from client.
- Easy to switch to different database.
- Presentation and Logic layers still tightly connected.
- Heavy load on server.
- Potential congestion on network.

3 - Tier Architecture

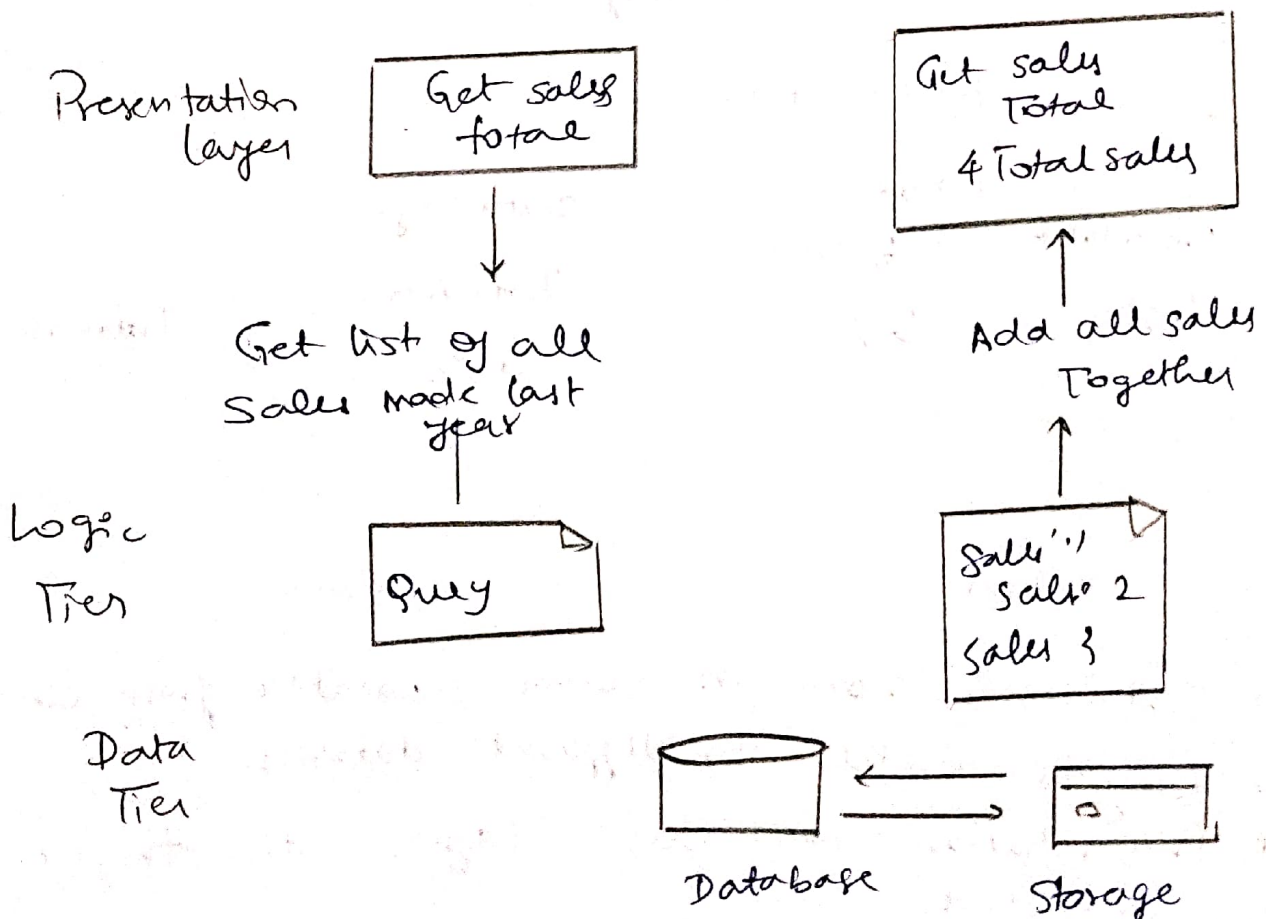


Client

Server

DB server

- Each layer can potentially run on different machine.
- Presentation layer, data layers disconnected.
- Three-tier Architecture is a client-server software architecture pattern in which the user interfaces functional logic, compute data storage and data.



Presentation Layer,

Static or dynamically generated content rendered by the browser (front-end)

Logic Layer,

A dynamic content processing and generation level application server. Eg: Java EE, ASP.NET, PHP etc

Data Layer:

A data base layer, comprising both sets and the database management system or RDBMS software that manages and provides access to the data.

Advantages:

- + Easier to maintain.
- + Components are reusable.
- + Faster development.
- + Web designer does presentation.
- + Software engineer does logic
- + DB admin does data model.